

B.Tech. Degree II Semester Regular Examination July 2024

EE/EC/CS/IT 23-200-0203B DIGITAL ELECTRONICS
(2023 Scheme)

Time: 3 Hours

Maximum Marks

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Understand the fundamental boolean functions and basic building blocks of digital systems.

CO2: Design optimal digital circuits using basic building blocks.

CO3: Analyse basic digital circuits.

CO4: Understand HDL models of simple circuits.

Bloom's Taxonomy Levels (BL): L1 - Remember, L2 - Understand, L3 - Apply, L4 - Analyze, L5 - Evaluate, L6 - Create

PO - Programme Outcome

PART A

(Answer ALL questions)

(5 × 2 = 10) Marks BL

- I. (a) Simplify the following Boolean expression using Boolean algebra. 2 L3
 - (i) $AB + \overline{A}C + A\overline{B}C(AB + C)$
 - (ii) $(X + \overline{Y} + XY)(X + \overline{Y})(\overline{X}Y)$
- (b) Find the minterm expansion of the following Boolean expression. 2 L3
 - (i) $F(A, B, C, D) = \overline{A}(\overline{B} + D) + AC\overline{D}$
 - (ii) $F(X, Y, Z) = X(\overline{X} + Y)(\overline{X} + Y + \overline{Z})$
- (c) Define the figure of merit of logic gates. The following are the voltage and current parameters of a 74 series logic gate, find fan out and figure of merit. 2 L3

Propagation delay = 9 ns,
Power dissipation per gate = 10 mW,
 $I_{OH} = 800 \mu A$; $I_{IH} = 40 \mu A$; $I_{OL} = 48 \text{ mA}$; $I_{IL} = 1.6 \text{ mA}$.
- (d) Draw the timing diagram showing the loading of a 4-bit input data 0101 into a 4-bit SISO register with initial state of each flip flop in the register being logic 1. 2 L
- (e) Draw the circuit of a 3-bit ring counter. Explain its working with timing diagram. 2 1

PART B

(4 × 10 = 40)

- I. (a) Determine the prime implicants of the Boolean function by using the tabulation method 6

$F(a, b, c, d) = \sum m(1, 4, 6, 7, 8, 9, 10, 11, 15)$.
 - (b) Solve using a 3 variable K map 4
 - (i) $AB\overline{E}\overline{C} + ABC\overline{D} + ABCE + A\overline{B}C\overline{D} + \overline{A}BC$
 - (ii) $m_1 + Dm_2 + m_3 + \overline{D}m_5 + m_7 + d(m_0 + Em_1 + \overline{E}m_6)$.
- OR**
- (a) Design a logic circuit to produce a HIGH output if and only if the input, represented by a 4-bit binary number, is greater than 12 or less than 3. Develop the truth table and draw the logic diagram. 6
 - (b) Calculate by 2's complement method 4
 - (i) $(75.75)_{10} - (126.250)_{10}$
 - (ii) $(A3.74)_{16} - (23.52)_8$.

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Marks BL

- IV. Describe the working of a 4-bit look ahead carry adder. Clearly show the derivation of the equations and implementation of hardware.

10 L2

OR

- V. (a) Implement the following function using 8×1 MUX
 $F = \sum m(0, 1, 2, 3, 4, 10, 11, 14, 15)$
 (b) Realize JK flipflop using SR flipflop.

5 L3

5 L2

- VI. (a) Design and setup a MOD-6 ripple up-counter using positive triggered JK flipflop. Show the timing diagram of the counter showing the glitches in the diagram.

5 L6

- (b) Design and setup a circuit to generate the following sequence by indirect method.

5 L6

$$\overline{1} \ 1 \ 0 \ 1 \ 1 \ 1 \ 0$$

OR

- VII. Design and setup a synchronous sequential counter using JK flipflop and avoid lock-out condition for the sequence.

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L6

$$4 \rightarrow 6 \rightarrow 7 \rightarrow 3 \rightarrow 1 \rightarrow 4 \dots$$

- VIII. (a) Implement the following function using PLA
 $X(a, b, c, d) = \sum m(0, 2, 6, 7, 8, 9, 12, 13, 14)$
 $Y(a, b, c, d) = \sum m(2, 3, 8, 9, 10, 12, 13)$
 $Z(a, b, c, d) = \sum m(1, 3, 4, 6, 9, 12, 14)$

5

- (b) Implement a full adder using PROM.

5

OR

- X. (a) Draw and explain the working of a two input TTL NAND totem-pole configuration.

5

- (b) Four 7-bit hamming codes are transmitted serially through a noisy channel. Detect and correct errors, if any, in the even parity hamming code.

5

$$1001001011100111101100011011$$

om's Taxonomy Levels

- 23.33%, L3 - 42.22% L6 - 34.45%.

$$1001001011100111101100011011$$

$$(2+4+2)$$

$$4+4+4+4$$